Introduction to Natural Language Processing

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Outline

- Introduction
- Natural Language Processing
- Transformers
- Attention Mechanism
- Decoder-only models
- Generation techniques
- Prompt Engineering

Introduzione

- Irene Siragusa
  - Information and Communication Technologies (PhD)
  - Computer Science Engineering (MSc)
  - Computer Science and Telecommunication Engineering

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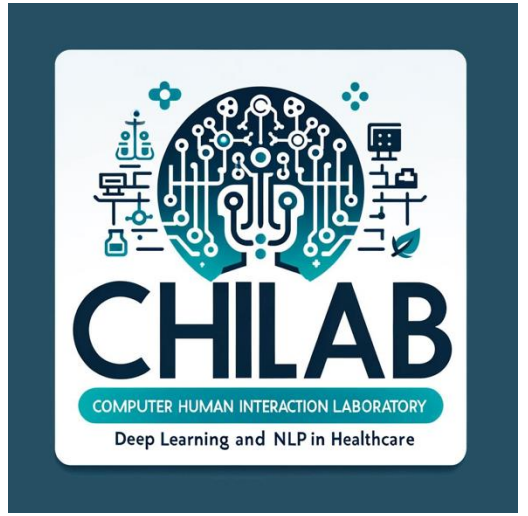
- Computer-Human Interaction Laboratory (CHILab) @ University of Palermo (UniPa) Palermo, Sicily, Italy



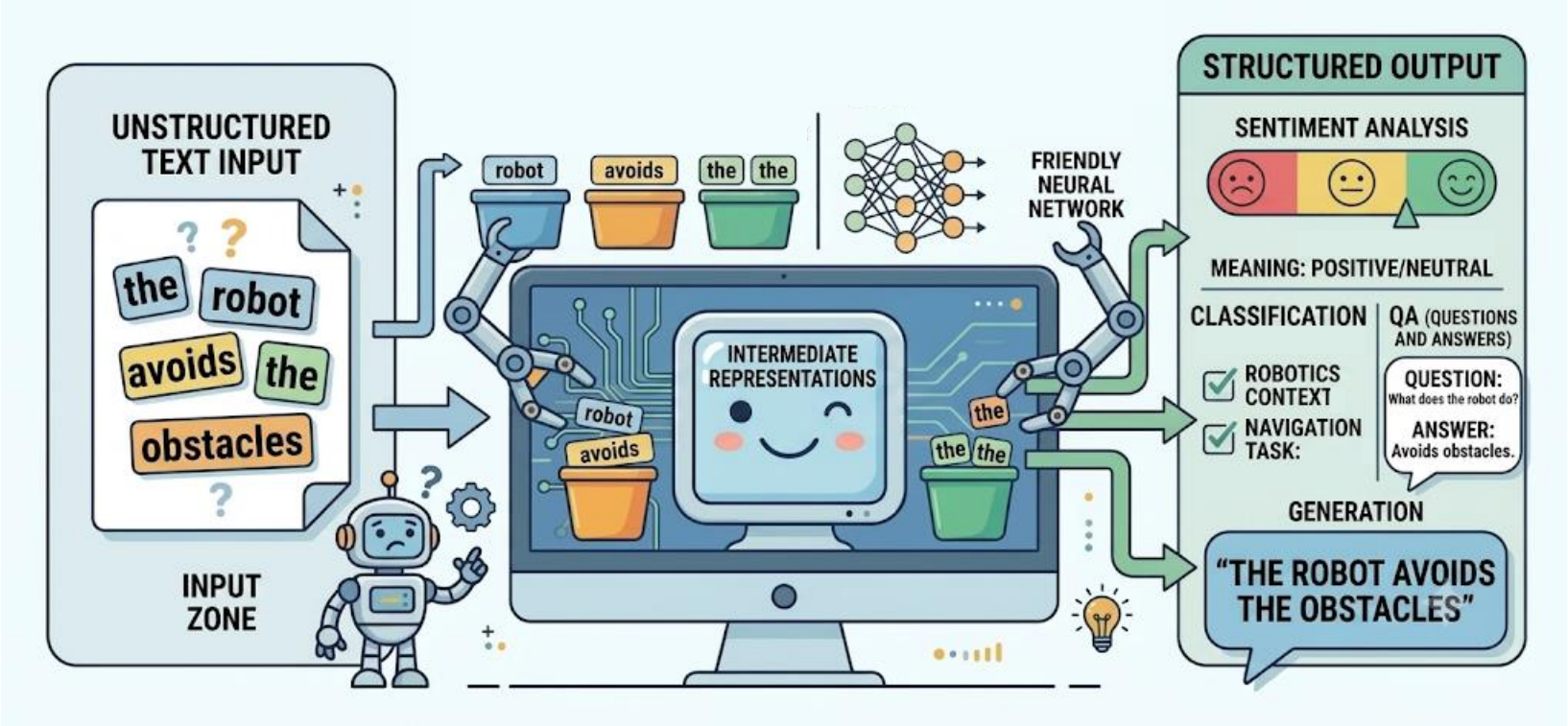
Introduzione

Subject material

- NLP-EN-JP GitHub
  - Slides with theory
  - python notebooks with practice

https://github.com/iresiragusa/NLP-EN-JP/

Natural Language Processing



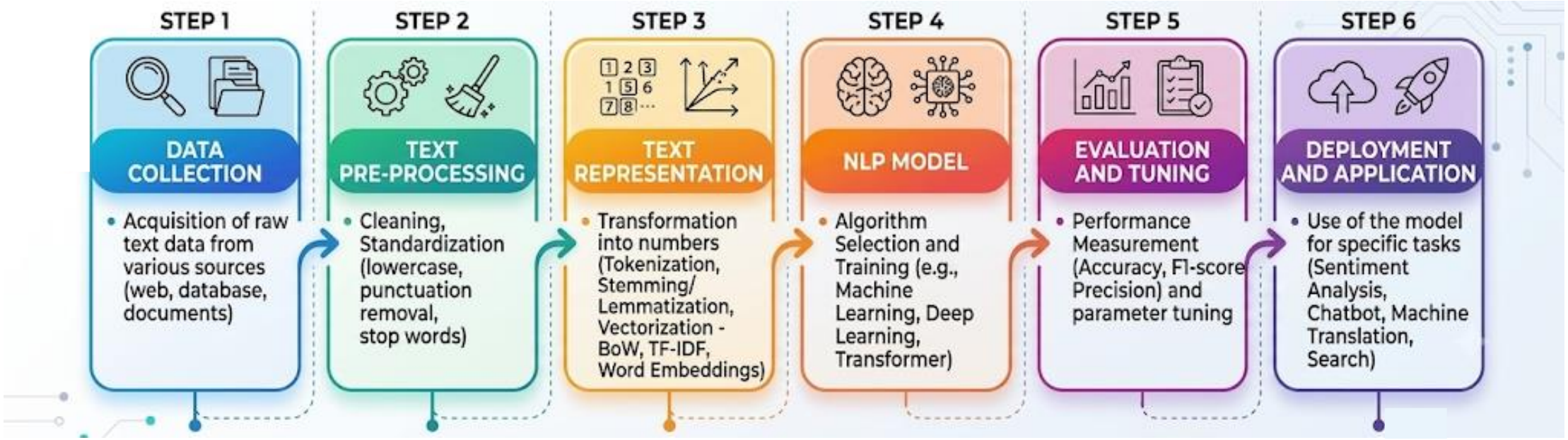
Natural Language Processing

- Natural Language Processing (NLP)
  - NLP is the use of human languages, such as English, Japanese or Italian, by a computer.
  - Many NLP applications are based on language models that define a probability distribution over sequences of words, characters, or bytes in a natural language.

NLP applications

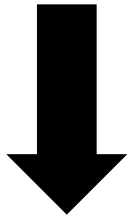
- Text pre-processing
- Information Extraction
- Machine translation
- Text Reasoning
- Summarization
- Textual classification
- Sentiment Analysis
- Hate Seech Detection
- Chatbots
- Text-to-Text Generation
- Text-to-Image and viceversa

NLP Pipeline



Tokenization

A misty ridge uprises from the surge



tokenization

['a', 'misty', 'ridge', 'up', '##rise', '##s', 'from', 'the', 'surge']

Large Language Models

- I Large Language Models (LLMs):

  ✅ Assign a probabability to given word sequences

  ✅ Generates text by sampling the most probable next words

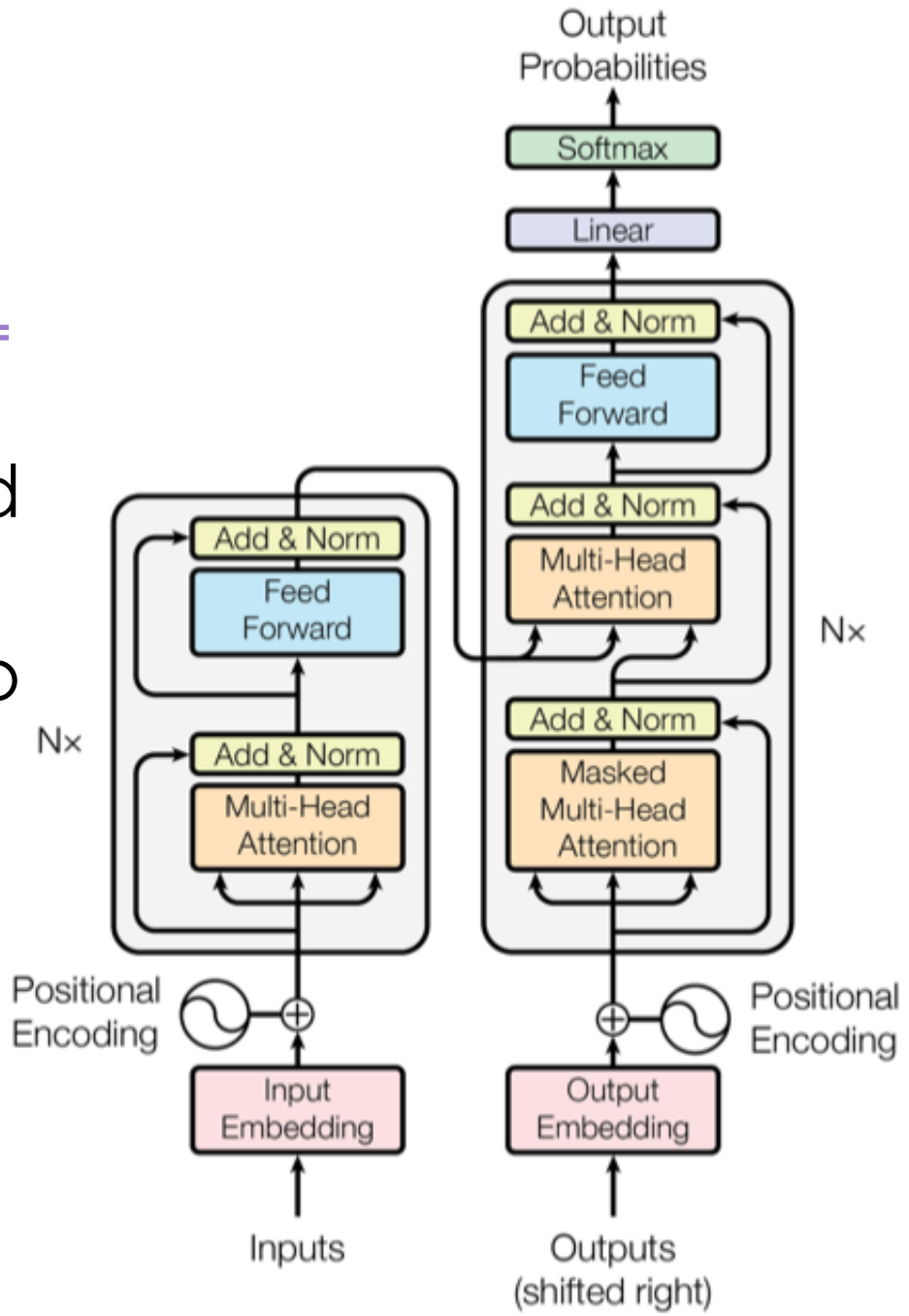
  ✅ Are based on transformers architecture

  💡 *Are trained to predict the next correct word, thus they do not have a proper knowledge*

Transformers

- Transformers are a Neural Network architecture made up of encoder and decoder
- They relies on attention mechanisms to draw draw global dependencies between input and output

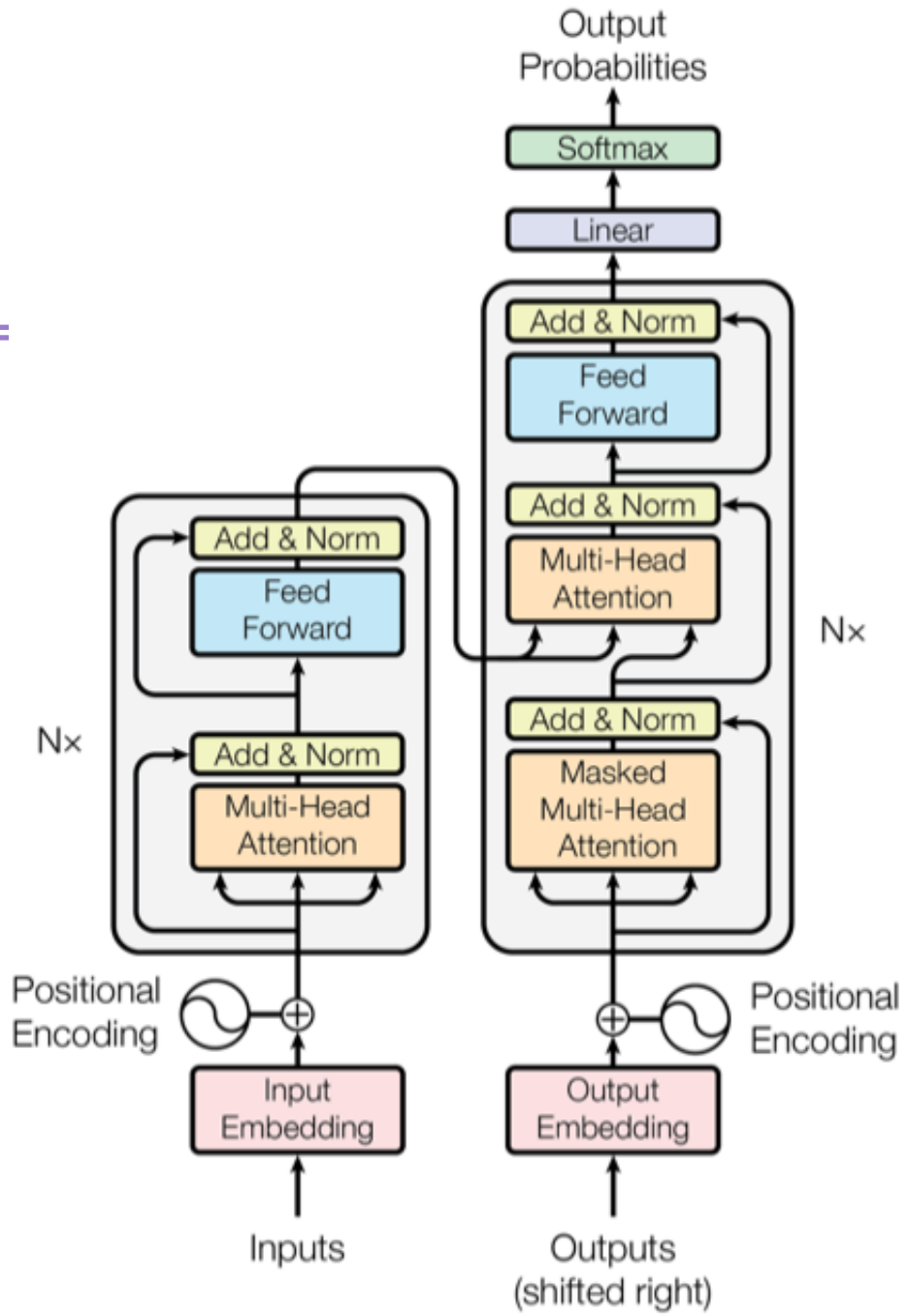
$$(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n) = \text{encoder}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$$
$$(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_m) = \text{decoder}(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n)$$



Vaswani A., Shazeer N., Parmar N., Uszkoreit J., Jones L., Gomez A. N. , Kaiser L., and Polosukhin I., Attention Is All You Need, http://arxiv.org/abs/1706.03762

Transformers

- Encoder
  - Multi-head self-attention mechanism + residual connection + normalization layer
  - Feed-forward fully connected layer + residual connection + normalization layer
- Decoder
  - Multi-head self-attention mechanism + residual connection + normalization layer
  - Multi-head attention over the output of the encoder stack + residual connection + normalization layer
  - Feed-forward fully connected layer + residual connection + normalization layer



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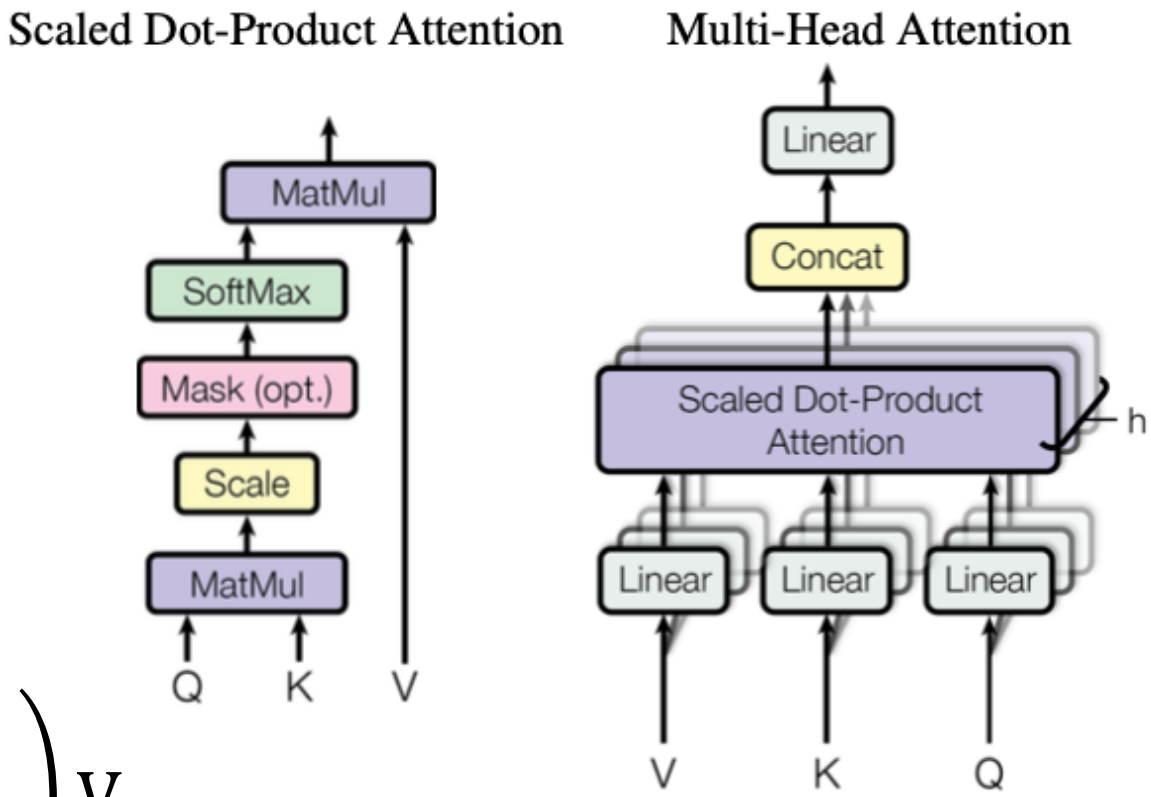
Transformers

From an input embedding three roles are played:

- Query → the current focus of attention
- Key → a preceding input
- Value → the value for the computation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\left(\sqrt{d_k}\right)}\right)\text{V}$$



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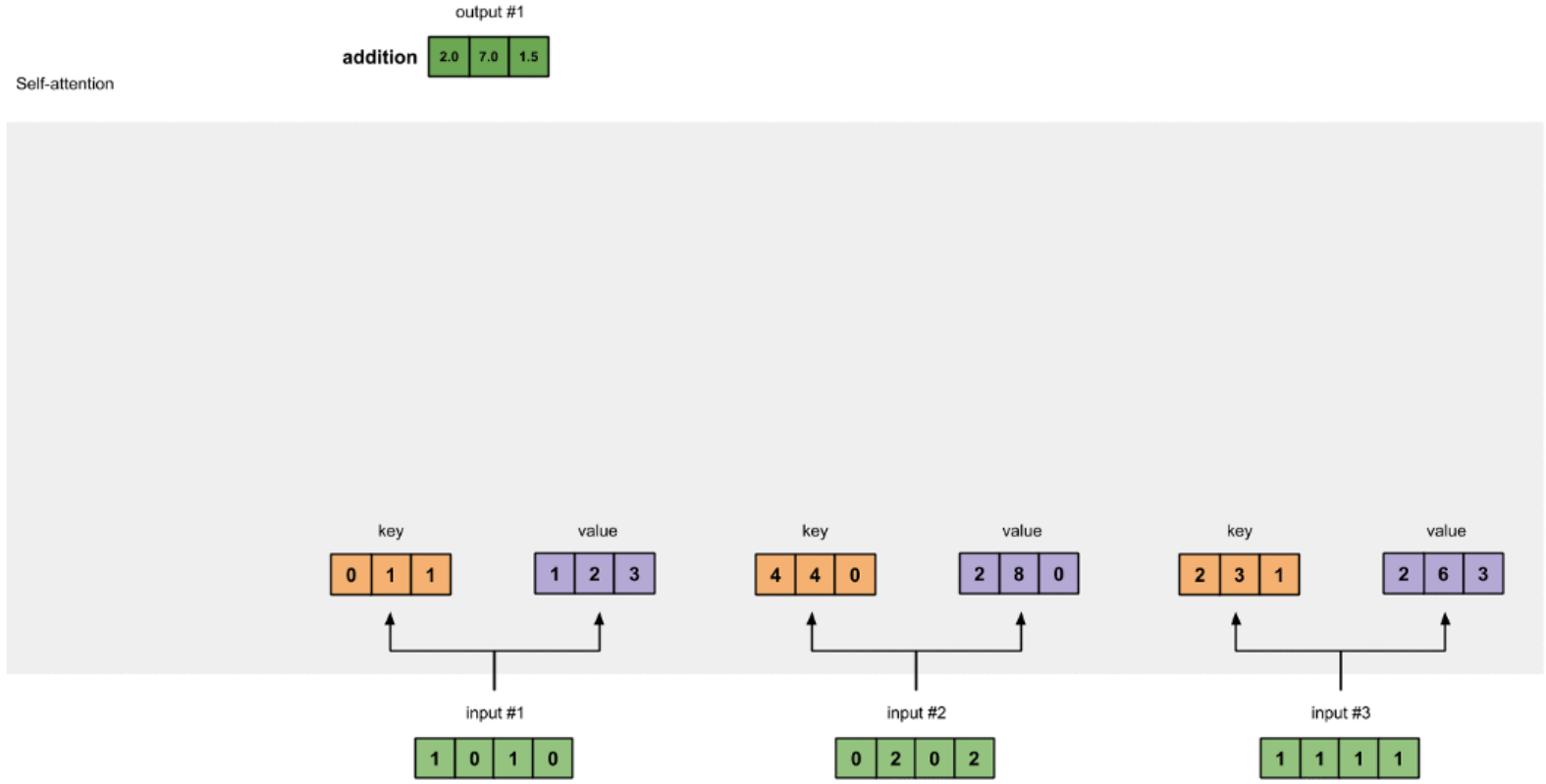
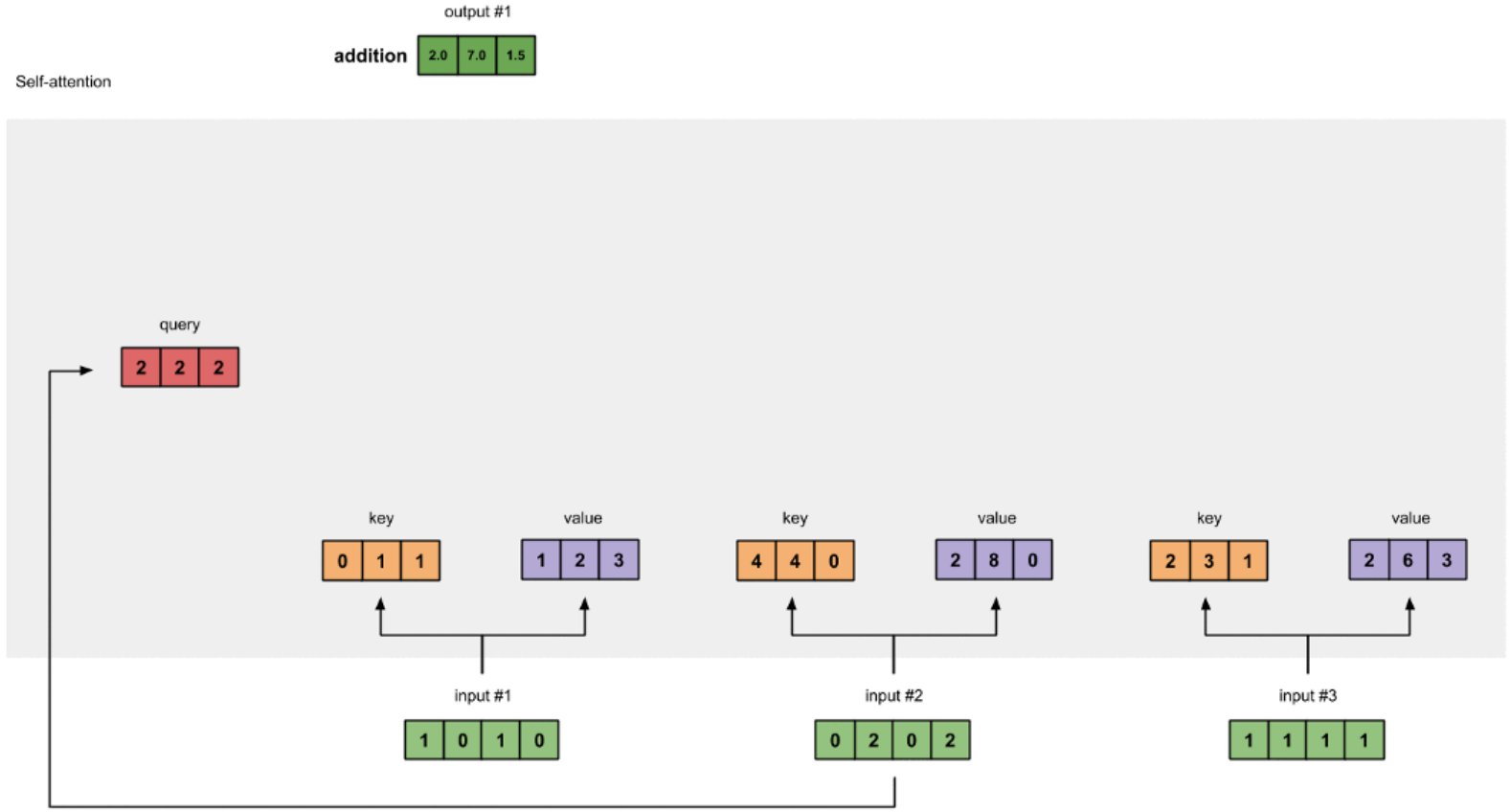


From a given input, calculate Key, Value

Transformers

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{(\sqrt{d_k})}\right)\text{V}$$



Calculate the Query of the given input

Transformers

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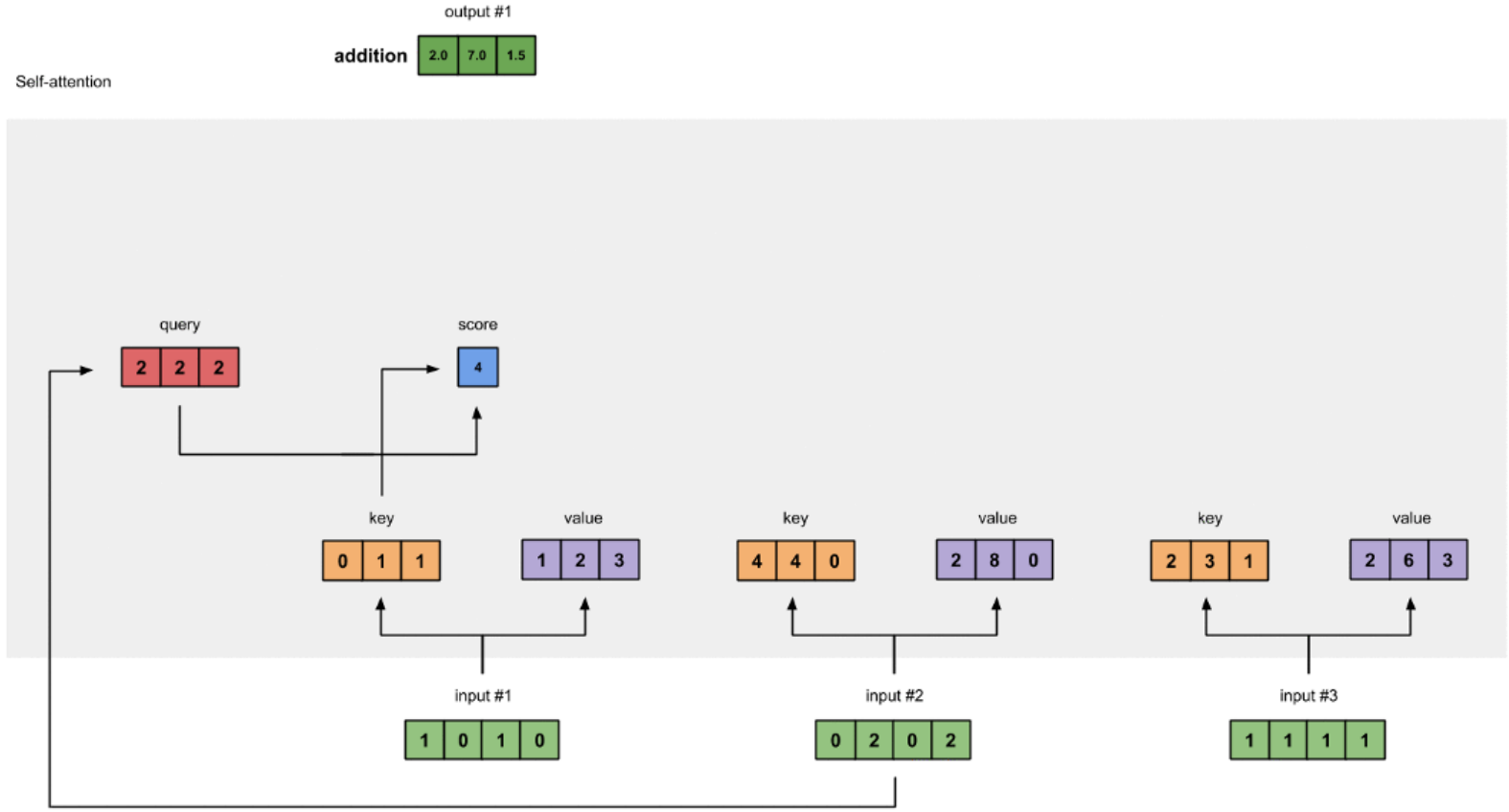
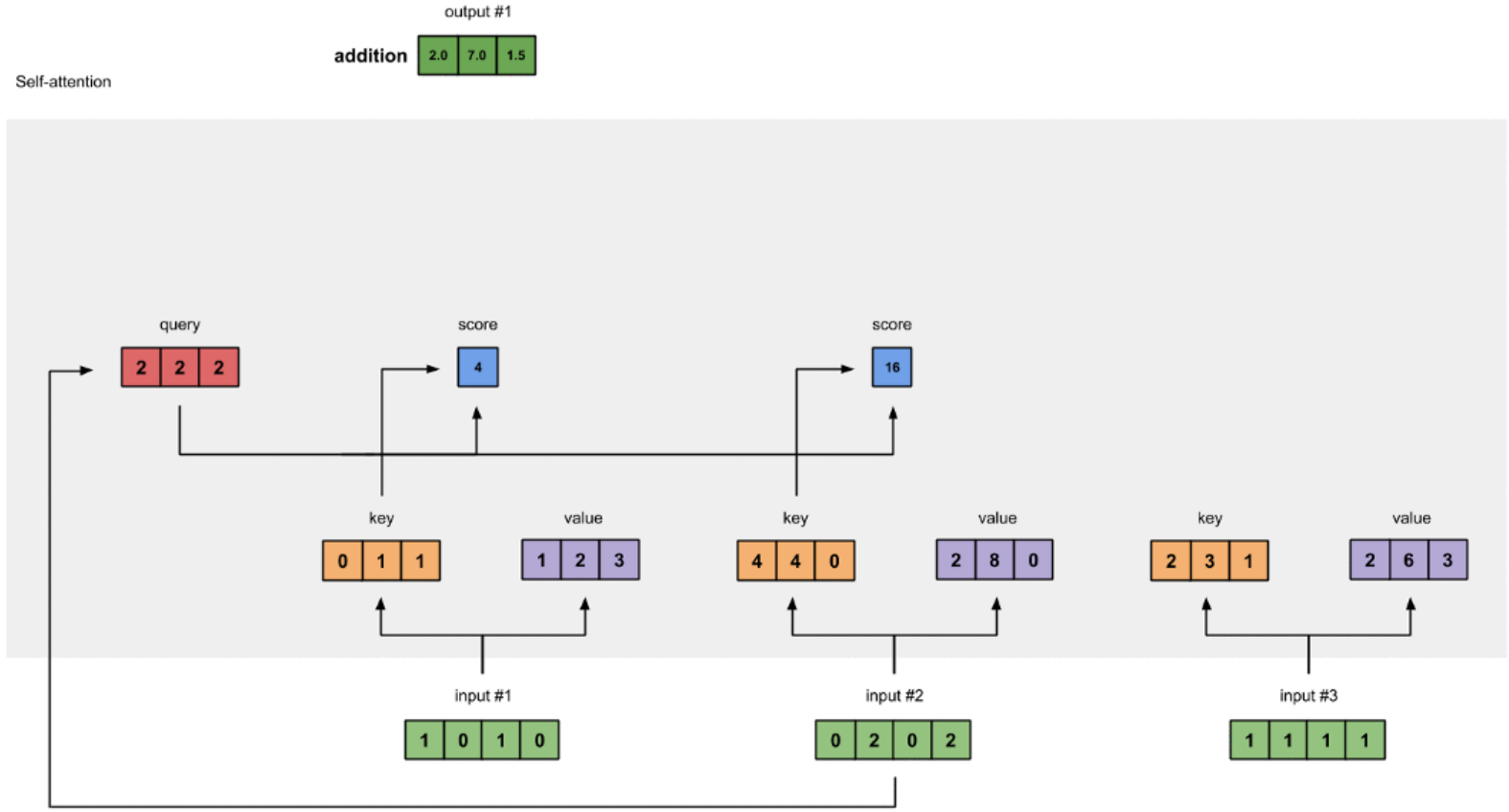


Calculate score as Query x Key, over alla the keys in the sequence

Transformers

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\left(\sqrt{d_k}\right)}\right)\text{V}$$



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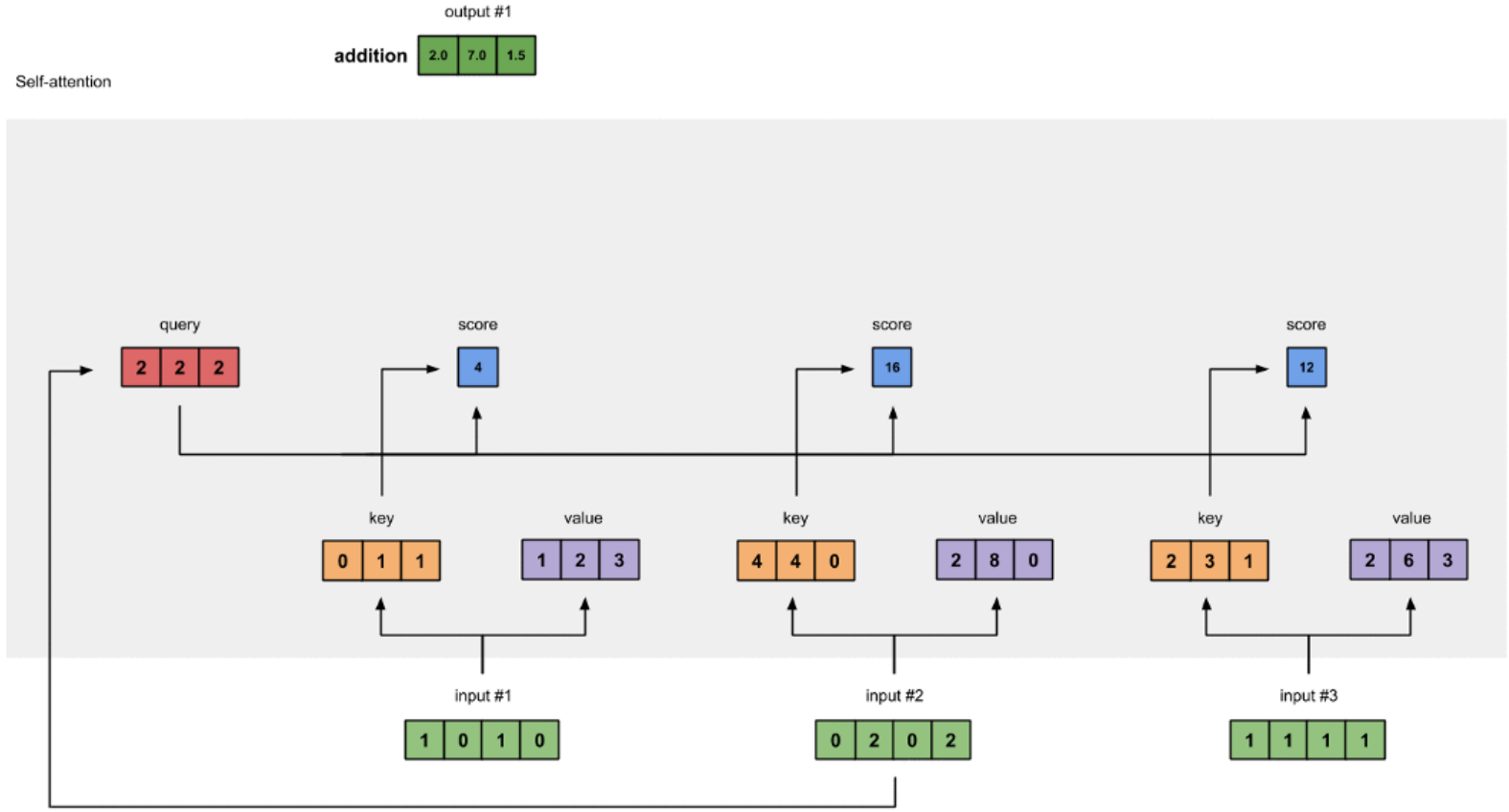

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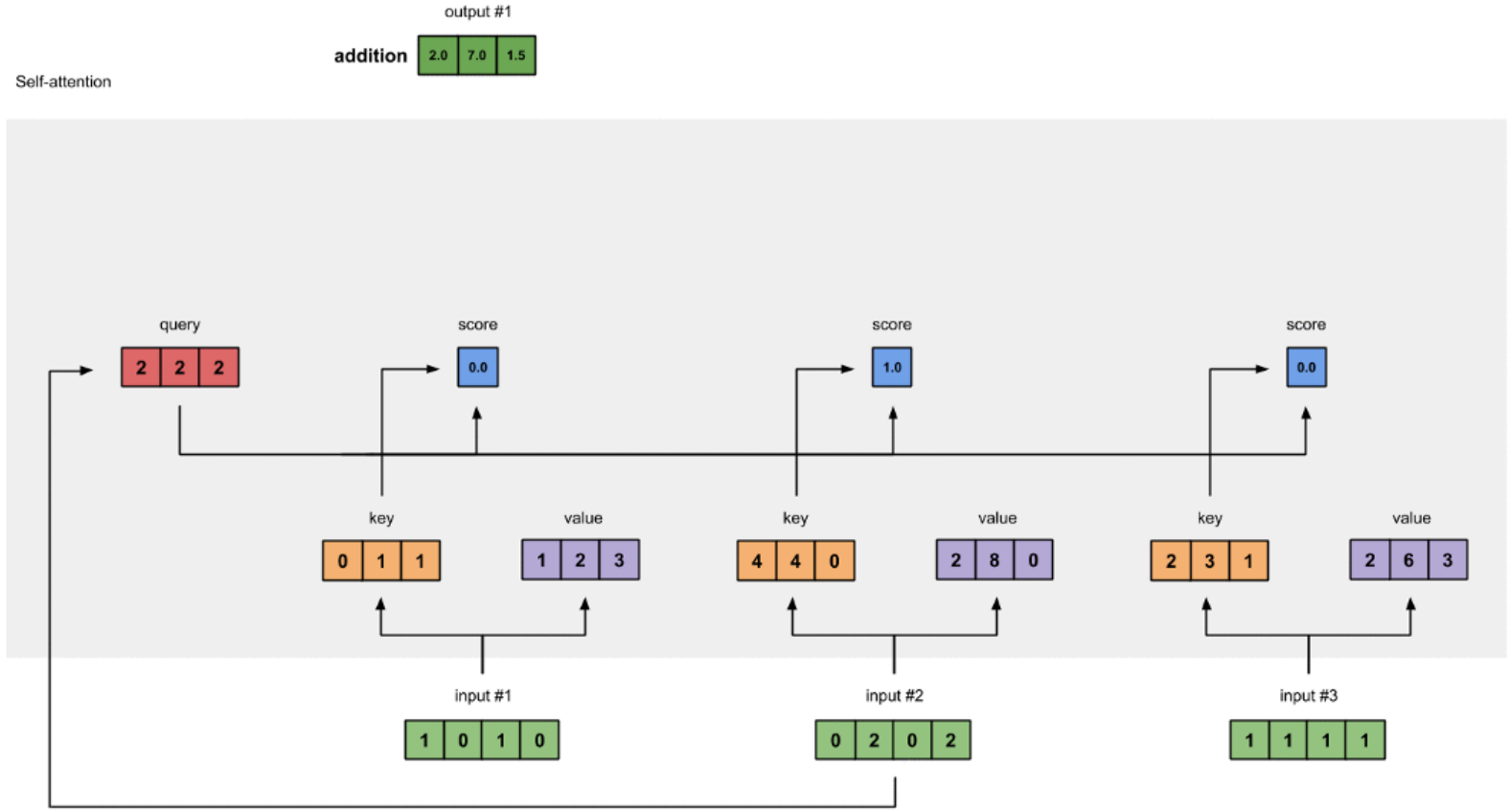
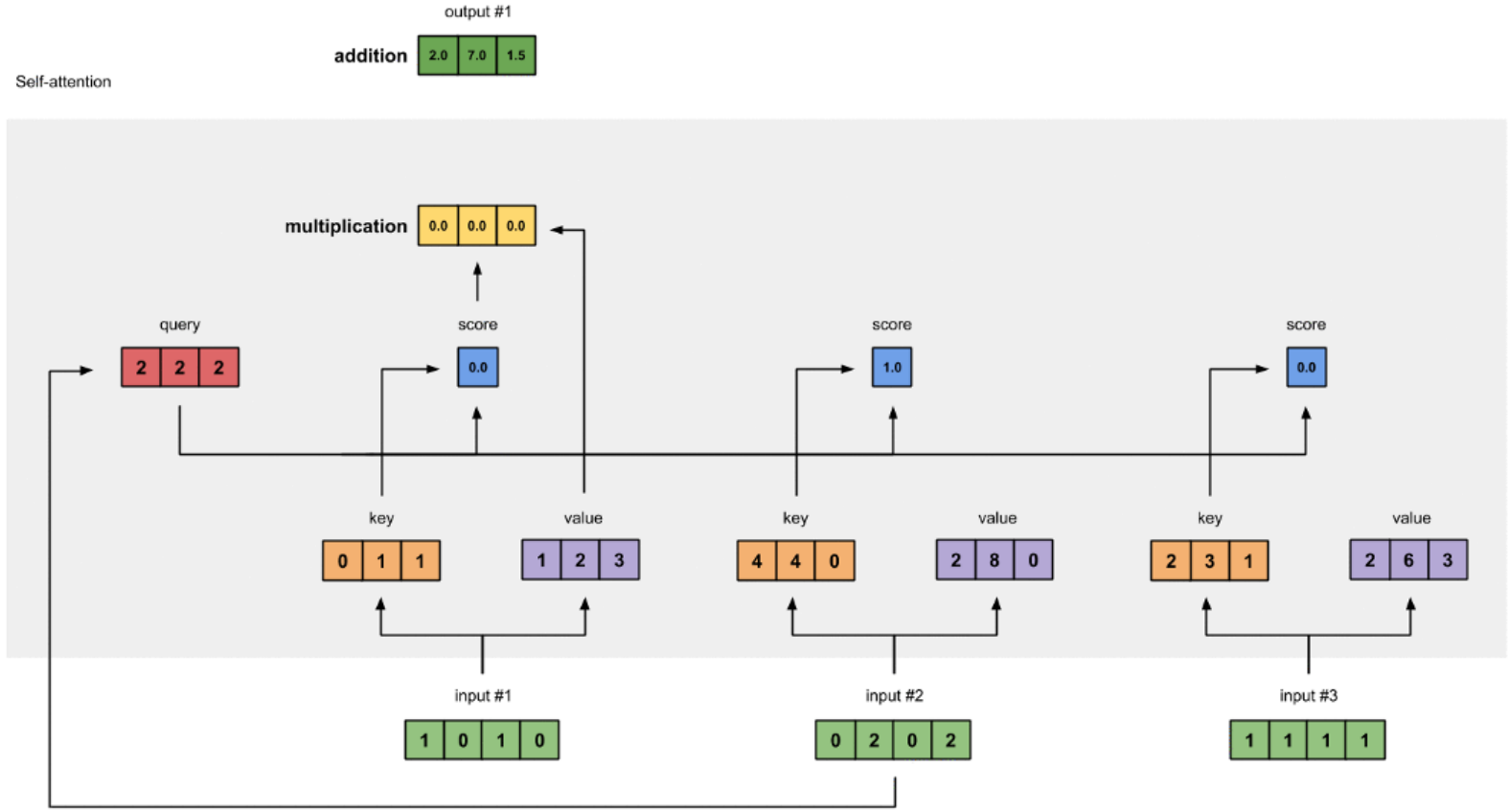


Apply softmax and transform the previous obtained scores in probabilities

Transformers

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\left(\sqrt{d_k}\right)}\right)\text{V}$$



Moltiply the obtained score for the Value to obtain the attention value associated to the input for the current Query

Transformers

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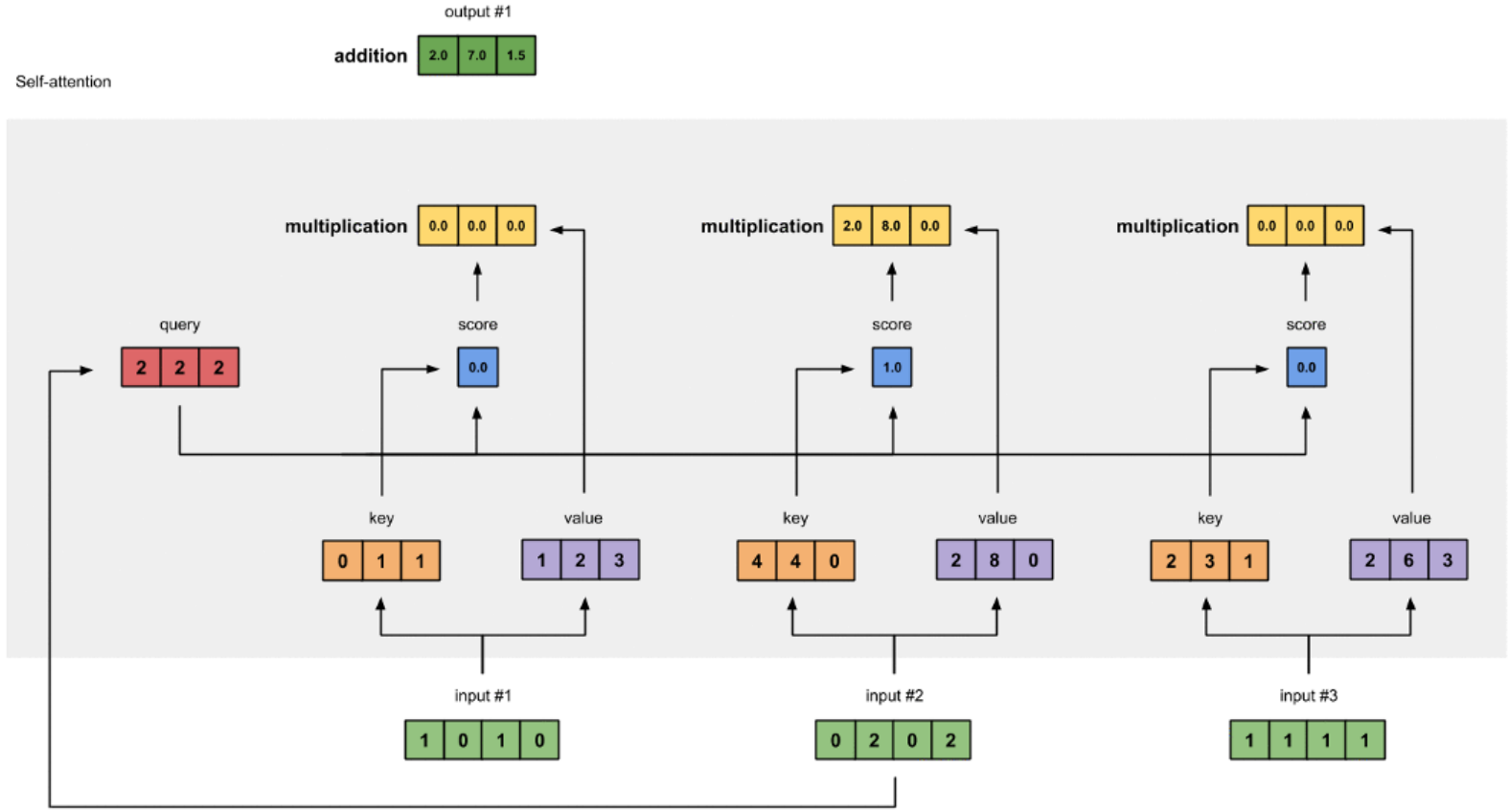

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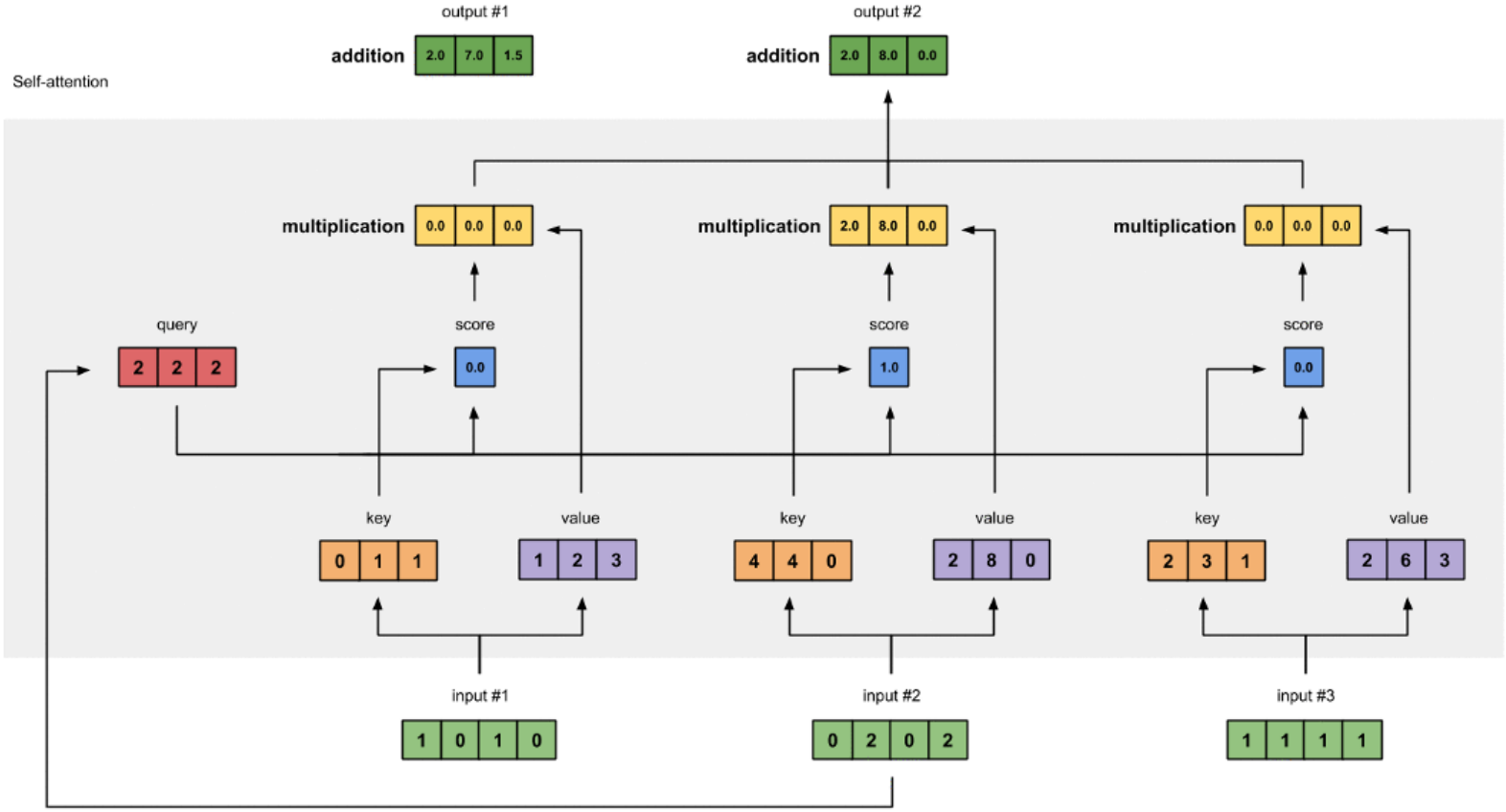
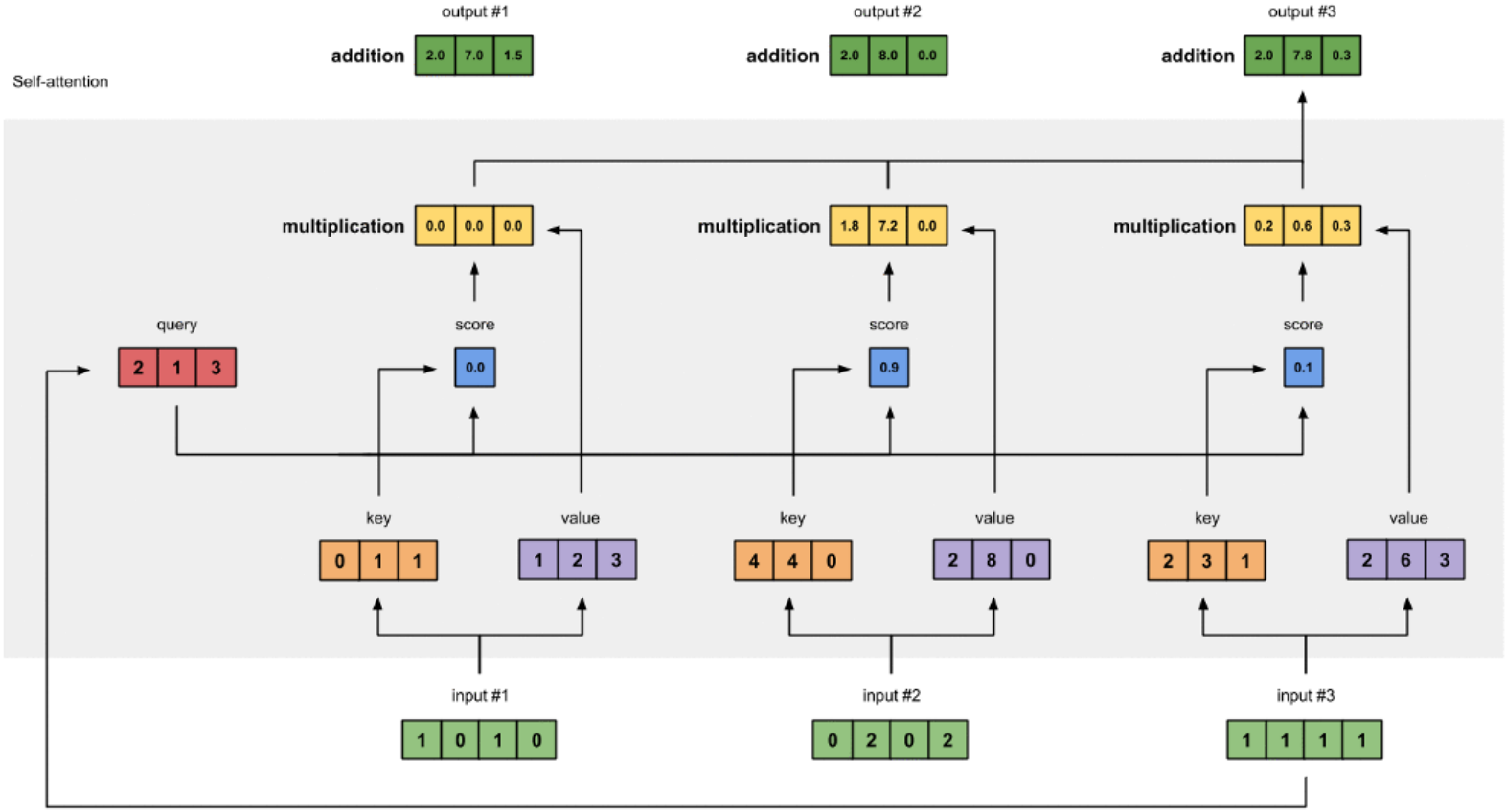


The output is the weighted sum of the obtained attention scores

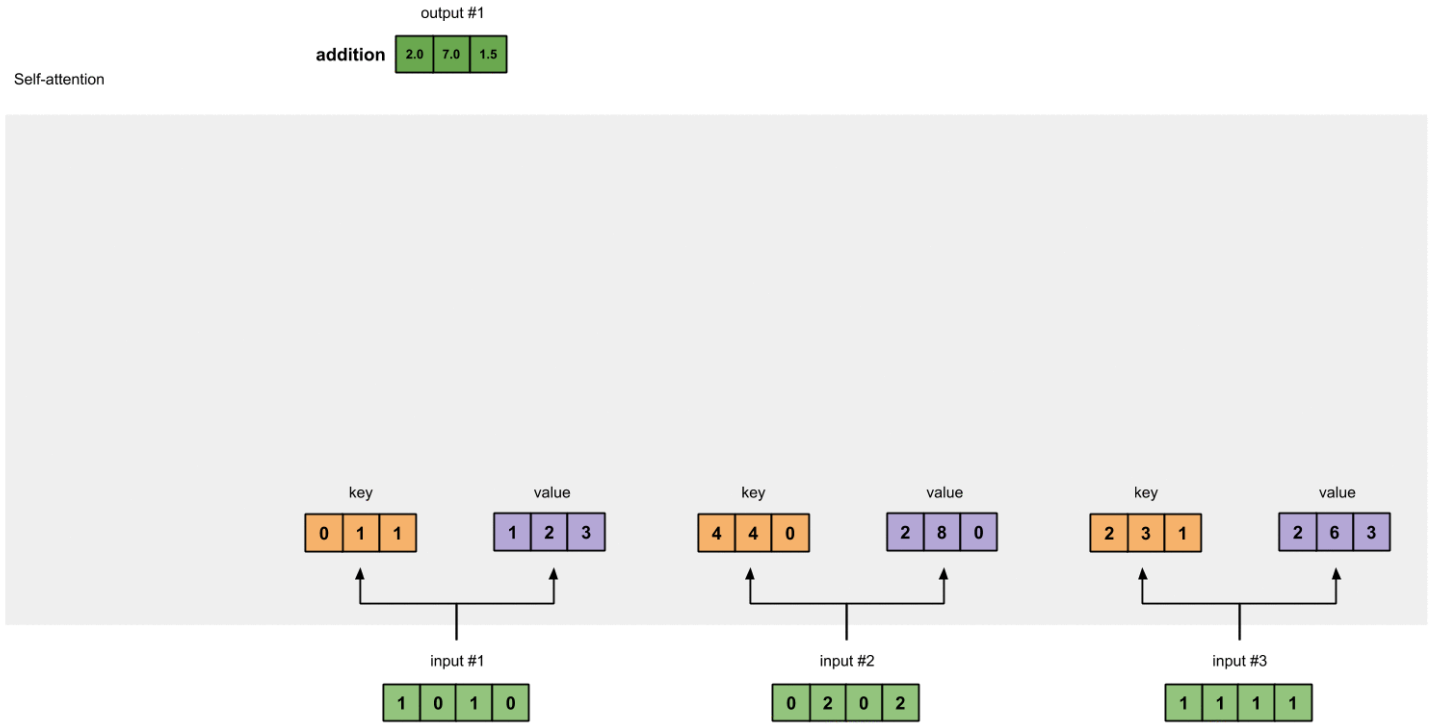
Transformers

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\left(\sqrt{d_k}\right)}\right)\text{V}$$

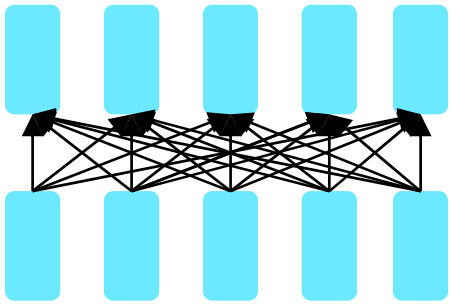


Iterate the process for all inputs in the sequence.
It is the self-attention mechanisms and I can see from left-to-right and from right-to-left

Transformers

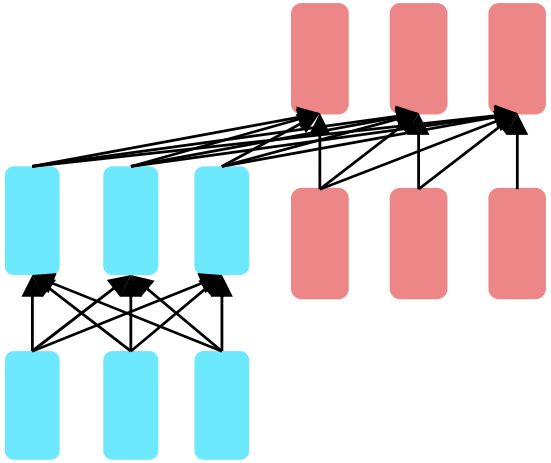


Large Language Models



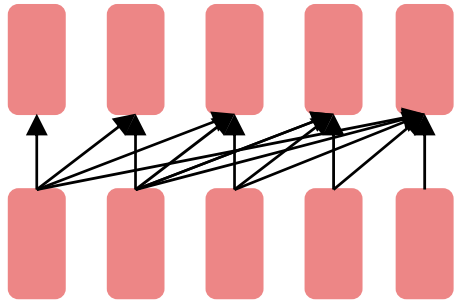
Encoders

BERT family



Encoder-decoders

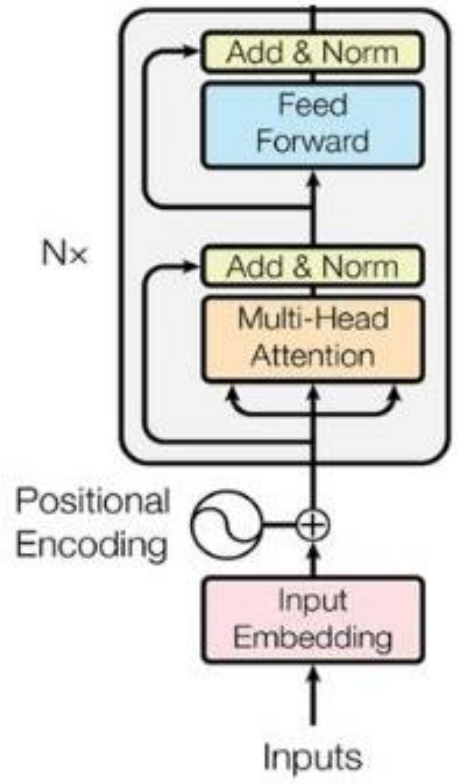
Flan-T5, Whisper



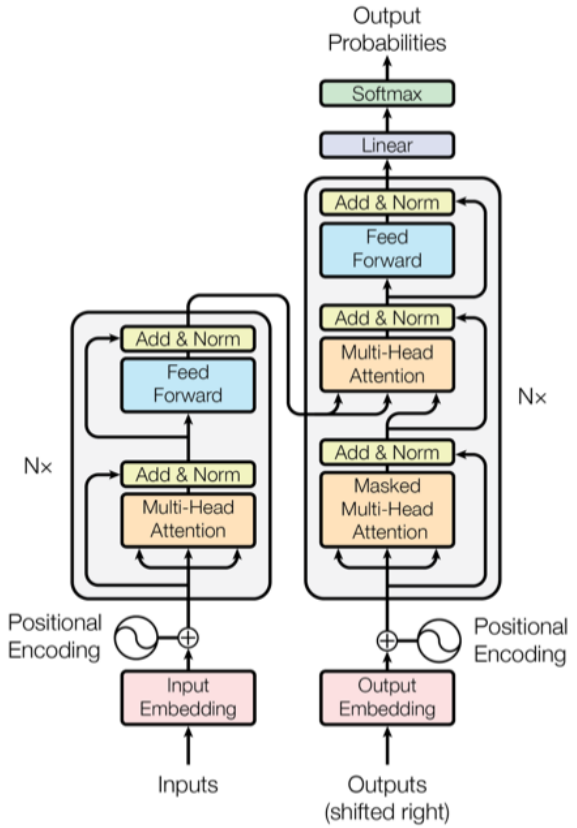
Decoders

GPT, Claude, Llama, Mixtral

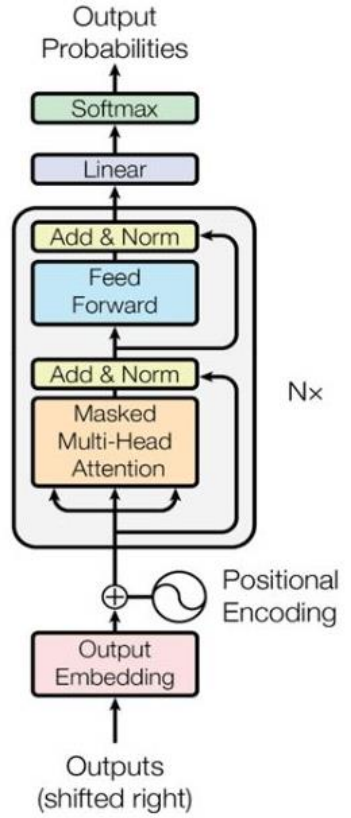
Large Language Models



Encoder-only

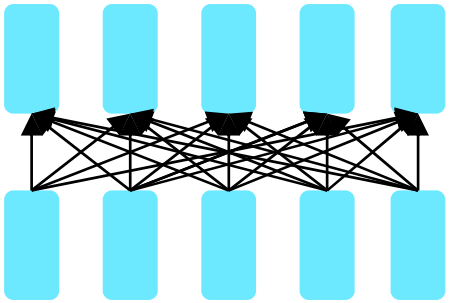


Encoder-decoder



Decoder-only

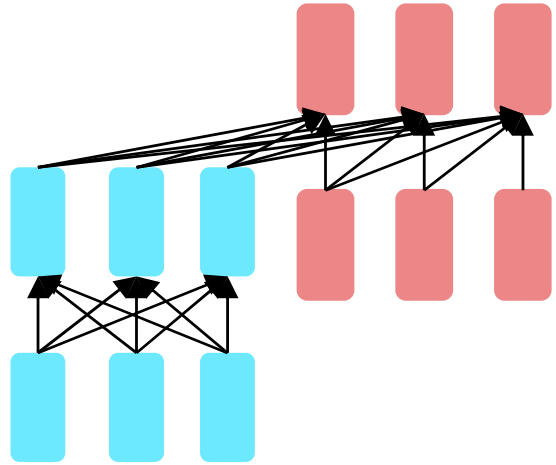
Encoder-only



- Masked Language Models (MLMs)
- BERT-like models
- Trained to predict a single masked word inside a given text
  - Fill the gap strategy
- Are usually finetuned (trained with supervised data) to execute specialized tasks s.a. classification

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

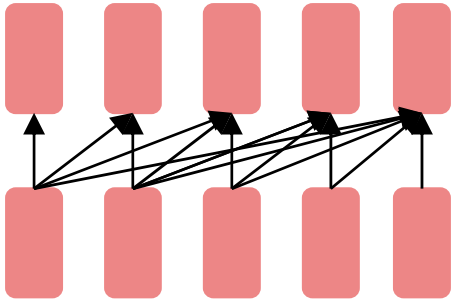
Encoder-Decoders



- Trained to map one sequence to another sequence
  - Machine Translation (mapping across languages)
  - Speech Recognition (mapping from acoustic features to words)

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

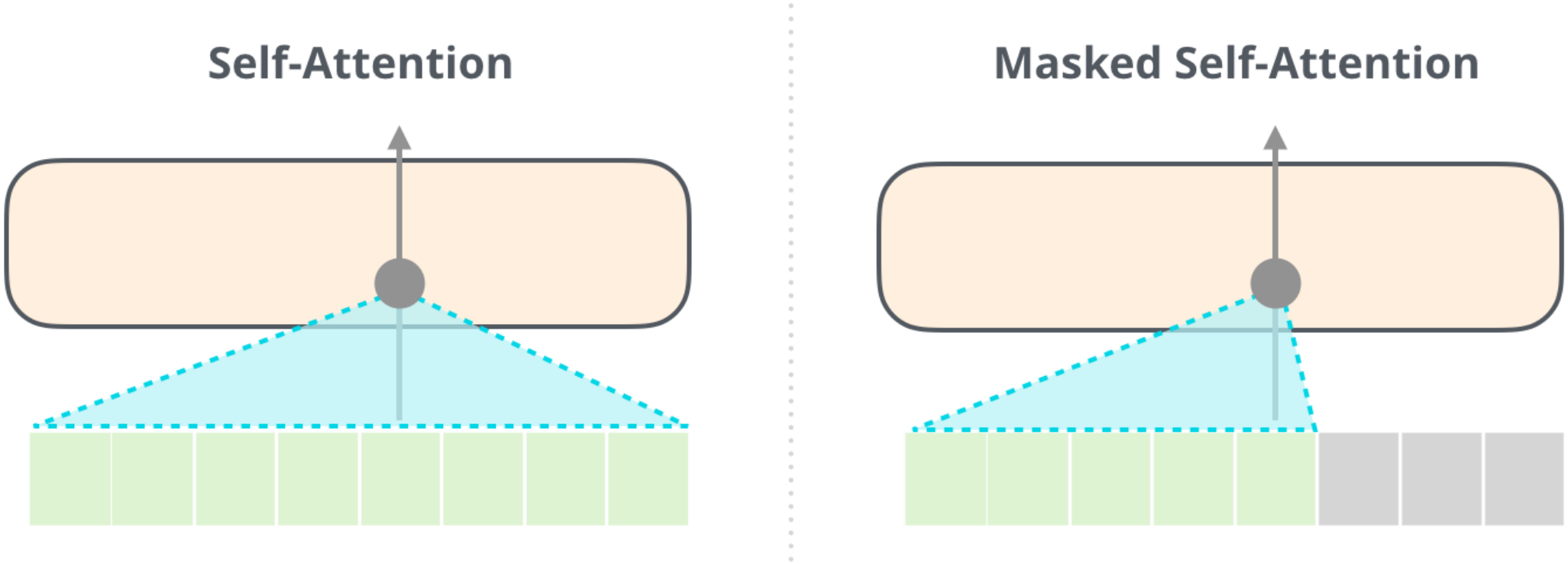
Decoder-only



- Causal LLMs, or Autoregressive LLM, or Left-to-right LLMs
- They are generative models, which can predict the next word from a given text
- All the previous tasks can be revised following a generative approach

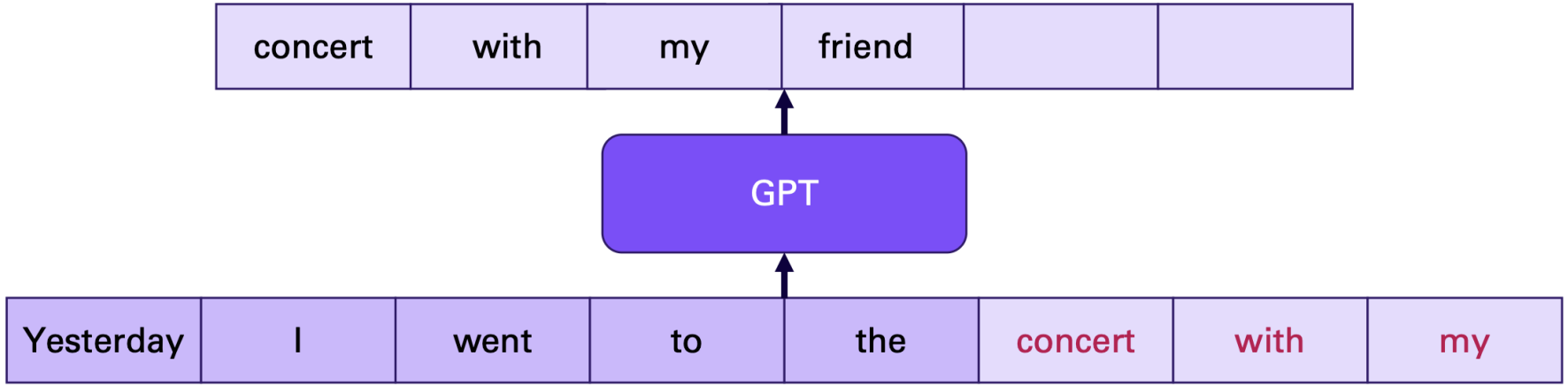
Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Decoder-only



Decoder-only – Next token prediction

- Masked self-attention + decoder-only
- Training technique to predict next token in a given sequence, the textual content is generated leveraging previous context



Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Decoder-only – Next token prediction

- Decoder-only models are generative models able to predict and generate newer words from the given input

- This can be simply formalized by the chain rule

$$P(w_{1:n}) = \prod_{i=1}^{n} P(w_i|w_{1:i-1})$$

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Decoder-only – Autoregression

- In training phase, the correct token is injected in the model, despite the predicted token which have been generated (teacher forcing strategy)
- In inference phase, the generated token is always injected into the model for subsequent generation (autoregression algoritm)
- If the output depends on previous tokens, I can define an *instruction*, called prompt, to be provided to the model, thus the desired text can be generated

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Decoder-only – Autoregression

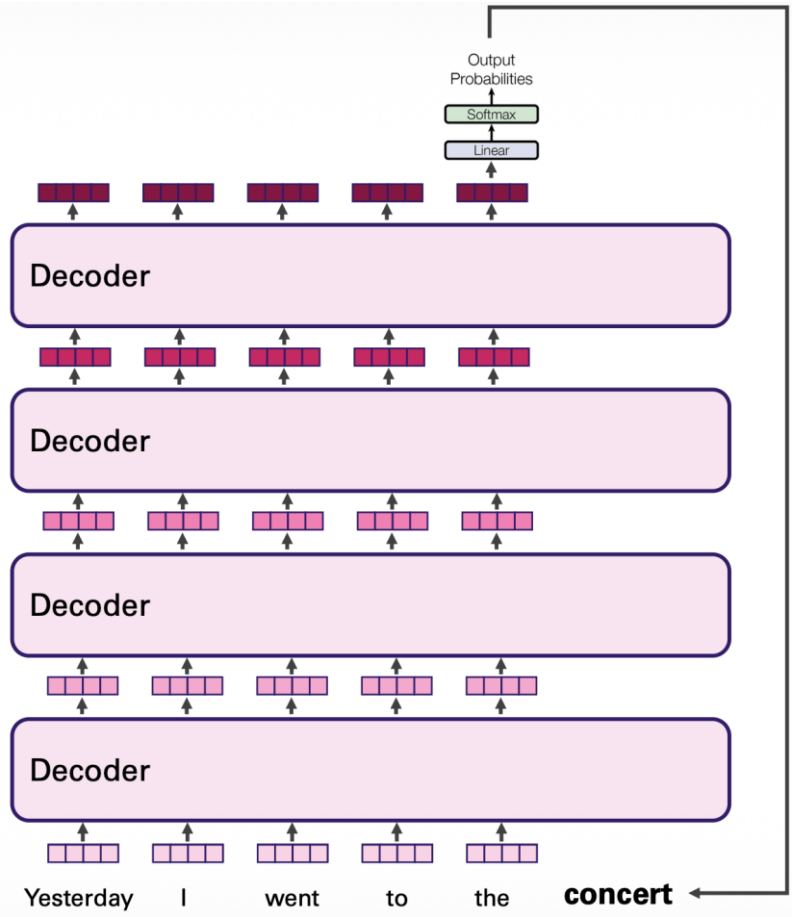
Algorithm 1 Overview of the autoregressive algorithm

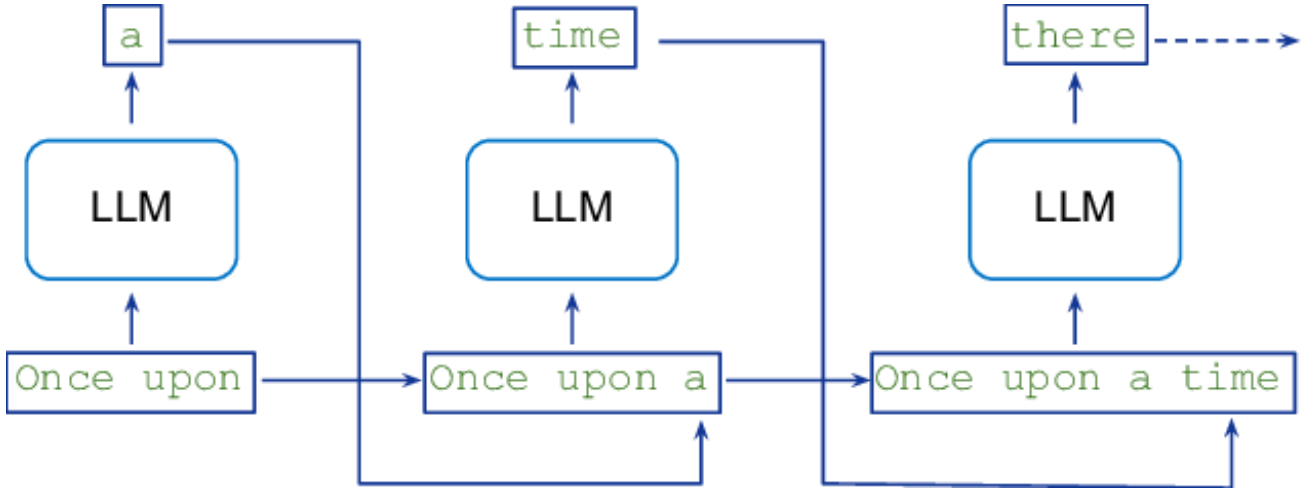
$i \leftarrow 1$
$w_i \sim P(w)$
while $w_i \neq \text{EOS}$ **do**
  $i \leftarrow i + 1$
  $w_i \sim P(w_i | w_{<i})$
end while

P(w) is the strategy over which tokens are predicted

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Decoder-only – Autoregression





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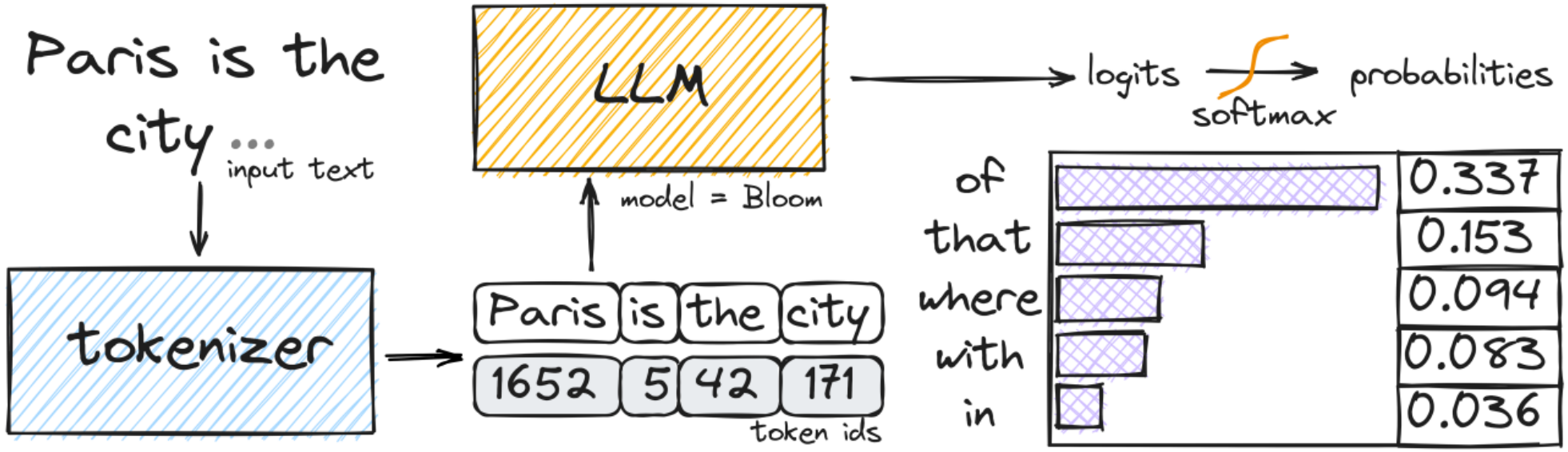
Decoder-only – Decoding

Text input → tokens → LLM → predicted token → predicted text



How do I select the output tokens? Decoding strategies

Decoder-only – Decoding

- Decoding is the selecting procedure of the next generated token, on the basis of the probabilities provided by the model itself

- Deterministic Decoding
  - Greedy search, beam search

- Stochastic Decoding (sampling)
  - Random sampling, Top-*k* sampling, Top-*p* sampling (nucleus sampling), Temperature sampling, contrastive seach

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Hands on:
E1 decoding strategies

Typology of LLMs

- Embedding LLMs
  - Encoder-only models
  - They generate vectorial representations of the given input text

- Generator LLMs
  - Decoder-only models
  - They generate natural language text from the given prompt

Typology of LLMs

- Foundational models
  - Base and general-purpose models
  - Trained over a huge quantity of data to capture a wide linguistic and contextual knowledge
  - Training with self-supervised techniques

- Instruction-tuned models
  - Models trained to follow the input-given instruction
  - Trained over annotated data sets
  - Training with supervised fine-tuning techniques (SFT)

Typology of LLMs

- Multilingual models
  - Base or instruct models with multi-lingual knowledge
  - They can be obtained from scratch or after a language adaptation via fine-tuning
- Multimodal models
  - Base or instruct models
  - Models which can process image, video, audio and textual content in input

Prompting

- Prompt: instruction given as an input to an LLM.
  - It creates a context able to guide the LLM in generating an output compliant with the user's objective
- Prompt Engineering: process to detect the optimal and effective prompt for a given task

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Prompting

Hotel Review Completions

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed.

Textual input from which we want a task to be executed

Prompting

Hotel Review Completions

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed. In short our stay was

Prompt for sentiment analysis

Prompting

Hotel Review Completions

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed. In short our stay was

---

... not a pleasant one. The staff at the front desk were not welcoming or friendly, and seemed disinterested in providing good customer service.

... uncomfortable and not worth the price we paid. We will not be returning to this hotel.

Negative text which affects the sentiment.

Prompting

- Prompts can be used for a huge variety of tasks

- Summarization
- Translation
- (Fine-grained) sentiment analysis
- …

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Templates

- We can rely on templates and reuse them for the same task, instead of re-write it everytime

Basic Prompt Templates

Summarization	{input} ; `tldr;`
Translation	{input} ; `translate to French:`
Sentiment	{input}; `Overall, it was`
Fine-Grained-Sentiment	{input}; `What aspects were important in this review?`

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Templates

- Prompts and templates should be designed in a non ambiguous manner, thus the desired output can be achieved.

A prompt consisting of a review plus an incomplete statement

Human: Do you think that "input" has negative or positive sentiment?
Choices:
(P) Positive
(N) Negative

Assistant: I believe the best answer is: (

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Templates

LLM Outputs for Basic Prompts

Original Review ($INPUT)	Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax and away from the city life.
Sentiment	**Prompt**: $INPUT + In short, our stay was **Output**: not enjoyable
Fine-grained Sentiment	**Prompt**: $INPUT + These aspects were important to the reviewer: **Output**: 1. Poor service 2. Unpleasant location 3. Noisy and busy area
Summarization	**Prompt**: $INPUT + tl;dr **Output**: I had a bad experience with the hotel's service and the location was loud and busy.
Translation	**Prompt**: $INPUT + Translate this to French **Output**: Je n'ai pas aimé le service qui m'a été offert lorsque je suis entré dans l'hôtel. Je n'ai également pas aimé la zone dans laquelle se trouvait l'hôtel. Trop de bruit et d'événements pour que je me sente détendu et loin de la vie citadine.

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Prompting with templates process

1. For a given task, design a suitable and specific template with a free input field.
2. Given the input and the template, the textual input is used to initialize the template (which now is the actual prompt) and given to the selected LLM.
3. With autoregressive decoding, the output token with the predicted answer are generated.
4. The output can be directly used as output or the answer can be extracted and/or refined from the generated content.

Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Few-shot prompting

- Improve a prompt by inclusion of labeled examples.
- On the contrary, zero-shots prompting uses prompts without any labeled samples.
- We can have *zero-, one-, n-shot prompting* o, in general, few-shot prompting

Few-shot prompting

Definition: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

Passage: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, Dangerously in Love (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Examples:

Q: In what city and state did Beyoncé grow up?
A: Houston, Texas

Q: What areas did Beyoncé compete in when she was growing up?
A: singing and dancing

Q: When did Beyoncé release Dangerously in Love?
A: 2003

Q: When did Beyoncé start becoming popular?
A:

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Few-shot prompting

How many examples?

- Non too many
- Higher increase in performance are obtained by first sample and decrease with the subsequents.
- Are useful to demonstrate the task to the LLM and the expected answer format.
- Samples can be both selected manually or automatically

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In-Context Learning

- All prompts can be seen as a learning process
  - The more LLM reads the prompt, the more its generation capabilities improves, mainly thanks to the given contextual information provided.

- This learning capability of generative LLM through prompt is referred to as in-context learning.

Chain-of-Thought Prompting

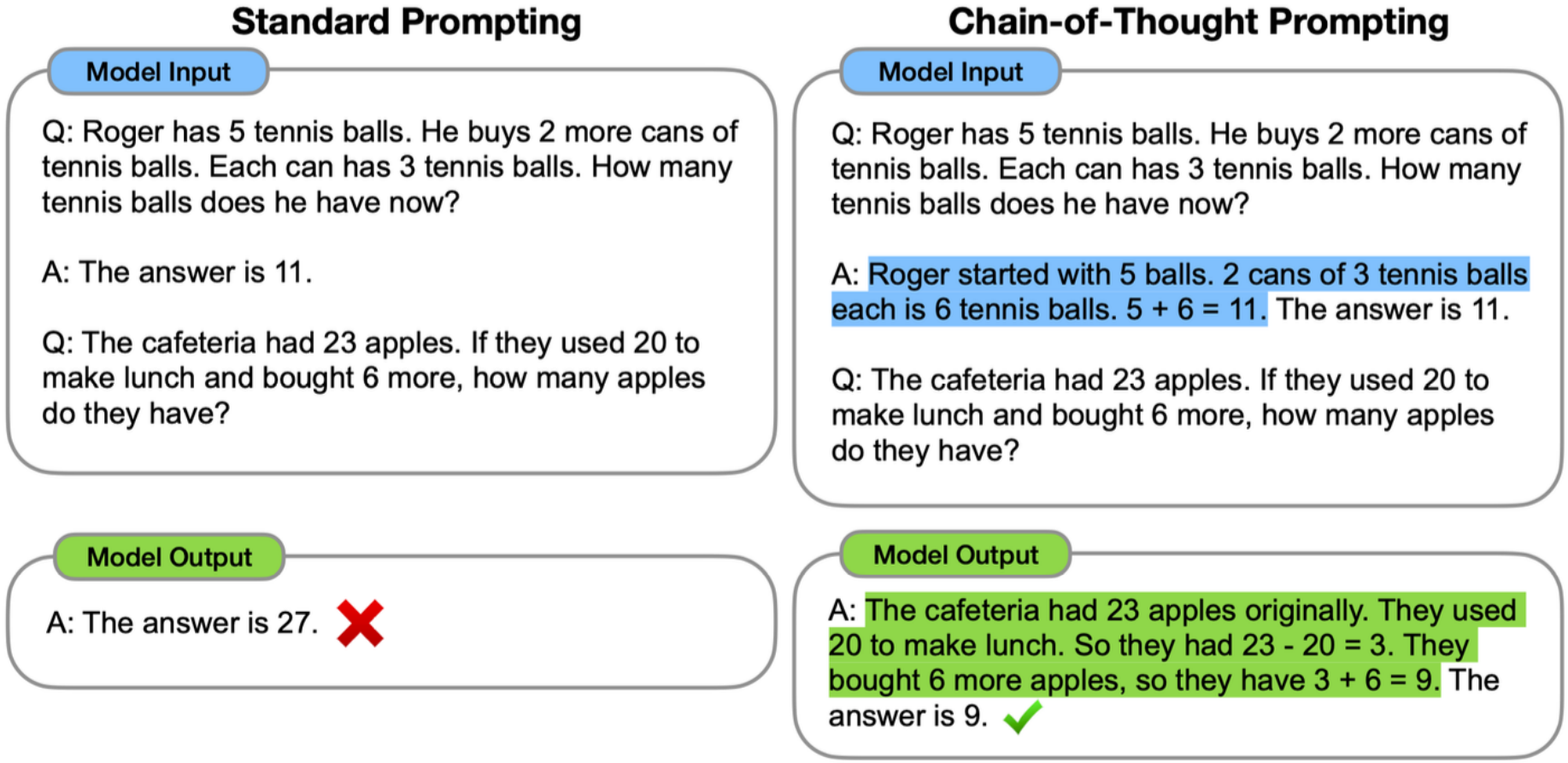
- Chain-of-Thought prompting (COT) is a technique to improve performances over difficult reasoning tasks over which LLMs tend to fails.
- People usually solve complex tasks by dividing it in simpler and smaller steps (divide et impera apporach).
- In COT, the instruction in the prompt forces the LLM to act in the same way.

Chain-of-Thought Prompting

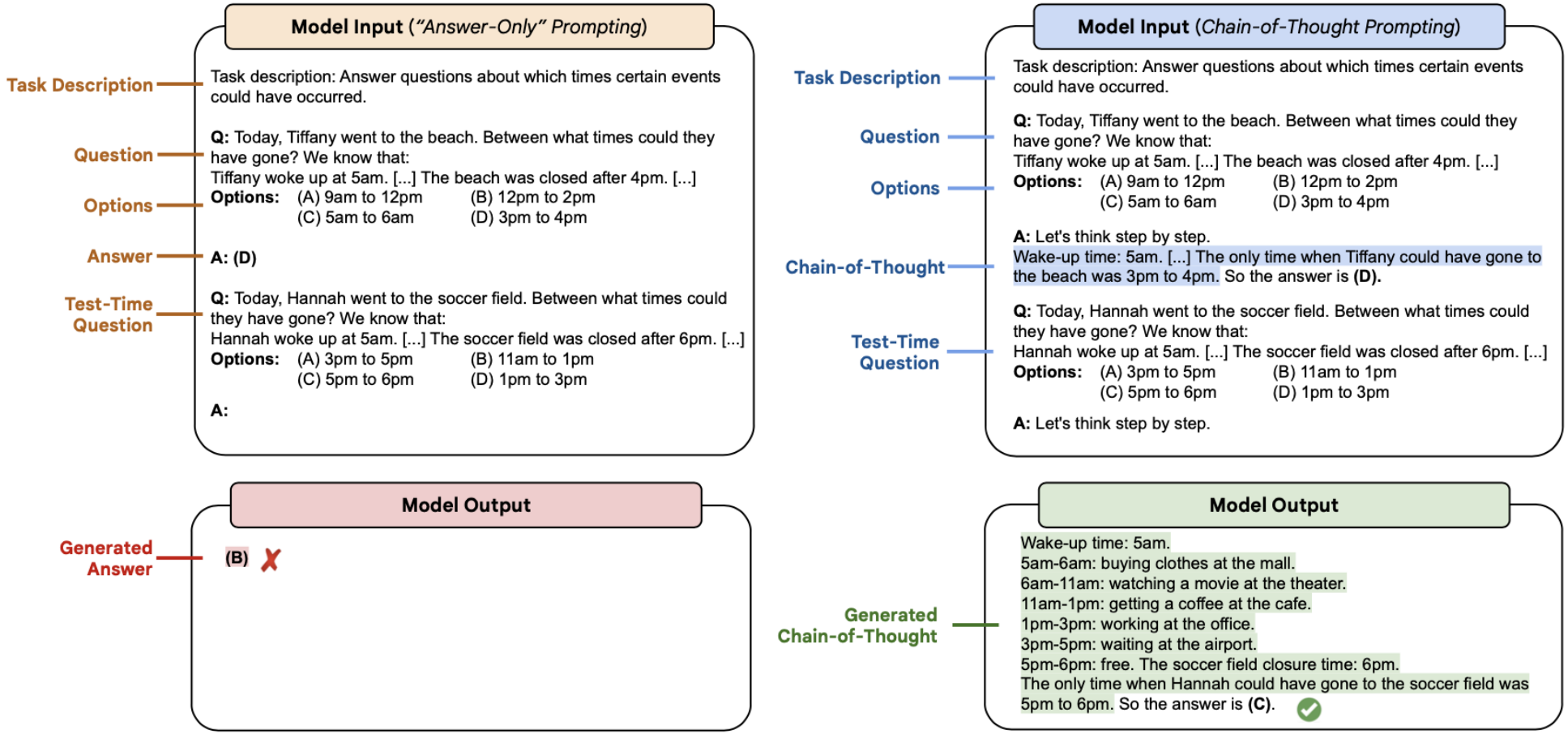
- Each sample in few-shot prompting is enriched with textual content which explains some reasoning steps, thus generating a logical demonstration.

- The objective is to force the LLM in producing analogous reasoning steps for the problem to be solved and in generating the correct answer as output.

Chain-of-Thought Prompting



Chain-of-Thought Prompting



open-source models @ HuggingFace

Gemma

Phi

Llama

DeepSeek

Mistral

Qwen

Libreries

- Model usage and inference
  - HuggingFace (Transformers)
  - vLLM
  - Ollama
- Complex pipelines
  - LangChain
  - LlamaIndex
- Training
  - Unsloth
  - HuggingFace (PEFT, Accelerate, TRL, bitsandbytes)

Bibliography

- Jurafsky D. and Martin J. H., Speech and Language Processing (3rd ed. draft), https://web.stanford.edu/~jurafsky/slp3/

Hands on:
E2 instruct inference